

EMCH 260

SOLID MECHANICS

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Combined Loadings

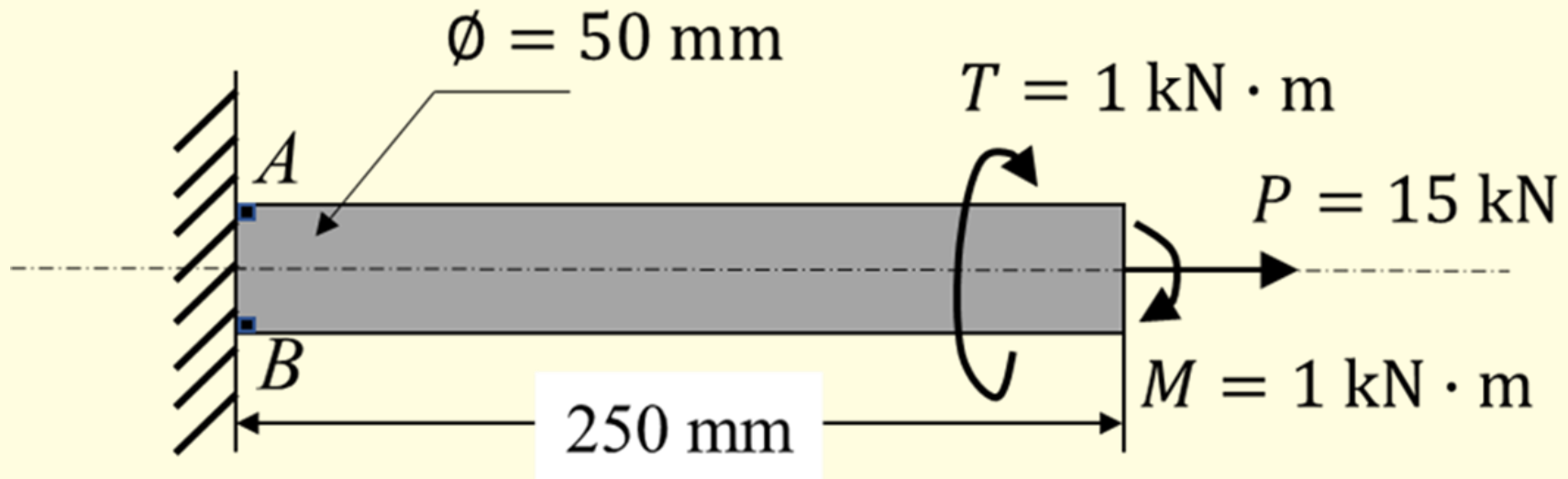
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Chapter 8.3

- ✓ Combined loading in cylindrical shafts including pressure vessels

IN-CLASS EXAMPLE

A cylindrical machine shaft (50 mm in diameter and 250 mm long) is supported at one end as a cantilever. The shaft is subjected to the combined loadings of tension, torque, and bending moment. Determine the normal and shear stress at point *A* (on the left-top surface) and point *B* (on the left-bottom surface).



IN-CLASS EXAMPLE, Cont'd

Solutions

At point A:

Both the bending moment and tensile force induces normal (**tensile**) stress

$$\sigma_A|_{\text{bending}} = \frac{M \cdot y}{I} = \frac{1000 \times 0.025}{\frac{\pi}{4} \times 0.025^4} = 81.49 \text{ MPa}$$

$$\sigma_A|_{\text{tensile force}} = \frac{P}{A} = \frac{15000}{\pi \times 0.025^2} = 7.64 \text{ MPa}$$

Then, the total normal stress at point A is

$$\begin{aligned}\sigma_A &= \sigma_A|_{\text{bending}} + \sigma_A|_{\text{tensile force}} = 81.49 + 7.64 \\ &= 89.13 \text{ MPa} \quad (\mathbf{Tensile}) \quad (\mathbf{Ans.})\end{aligned}$$

The external torque induces **shear** stress at point A:

$$\tau_A = \frac{Tc}{J} = \frac{1000 \times 0.025}{\frac{\pi}{2} \times 0.025^4} = 40.74 \text{ MPa} \quad (\mathbf{Ans.})$$

IN-CLASS EXAMPLE, Cont'd

Solutions

At point B:

The bending moment normal **compressive** stress, while tensile force induces normal **tensile** stress

$$\sigma_B|_{\text{bending}} = -\frac{M \cdot y}{I} = -\frac{1000 \times 0.025}{\frac{\pi}{4} \times 0.025^4} = -81.49 \text{ MPa}$$

$$\sigma_B|_{\text{tensile force}} = \frac{P}{A} = \frac{15000}{\pi \times 0.025^2} = 7.64 \text{ MPa}$$

Then, the total normal stress at point B is

$$\begin{aligned}\sigma_B &= \sigma_B|_{\text{bending}} + \sigma_B|_{\text{tensile force}} = -81.49 + 7.64 \\ &= -73.85 \text{ MPa} \quad (\mathbf{Compressive}) \quad (\mathbf{Ans.})\end{aligned}$$

The external torque induces **shear** stress at point B:

$$\tau_B = \frac{Tc}{J} = \frac{1000 \times 0.025}{\frac{\pi}{2} \times 0.025^4} = 40.74 \text{ MPa} \quad (\mathbf{Ans.})$$

OUT-OF-CLASS READING

- **Textbook, pp. 413 – 416**
 - **In particular, Example 8.1 (pp. 416)**